MIL-P-18703(NOrd)

2 June 1955

SUPERSEDING
NAVORD OS 3835B
13 November 1953
NAVORD OS 1223A
17 May 1944

MILITARY SPECIFICATION

PRIMER, COMBINATION, MARK 13 MODS 0, 1 AND 2

1. SCOPE

1.1 This specification governs the manufacture, assembly, loading and preparation for delivery of Primer, Combination, Mk 13 Mods 0, 1, and 2 and the methods of inspection and tests upon which acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

BUREAU OF ORDNANCE

NAVORD OS 919 -- Priming Caps.

STANDARDS

MILITARY

MIL-STD-105	--	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	--	Marking for Shipment and Storage.
MIL-STD-353	--	Transportation Vibration Test for Use in Production of Fuzes.

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DRAWINGS

BUREAU OF ORDNANCE

LD No. 90656 -- Case Ignition Combination Primer Mk 13 Mod 0,
LD 109468 -- Case Ignition Combination Primer Mk 13 Mod 1, List of Drawings,
LD 109510 -- Case Ignition Combination Primer Mk 13 Mod 2, List of Drawings,
LD 166537 -- Test Set (Primer) Mk 135, Mod 0,
LD 168789 -- Primer Container Mk 7 Mod 0,
LD 257521 -- Primer Container Mk 7 Mod 1,
563645 -- Primer Container Mk 11 Type,
and all documents listed on the above LDs and drawing.

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400 -- General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment

NAVORD OD 5823 -- Sensitivity Test of Primers and Detonators Using Test Set Mk 135 Mod 0 (Primer) and Mk 136 Mod 0 (Detonator).

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the Procuring Activity or as directed by the Contracting Officer.)

2.2 Other publications. -- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78 -- Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles.

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(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition-Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publications should cite "49 CFR 71-78 (Rev. 1950).")

3. REQUIREMENTS

3.1 General.

3.1.1 The Primers, Combination, Mk 13 Mods 0, 1, and 2 covered by this specification shall be manufactured in accordance with Bureau of Ordnance List of Drawings LD No. 90656, LD No. 109468, and LD No. 109510, respectively, and all documents listed thereon.

3.1.2 The requirements and procedures established in OP 400 shall be adhered to wherever applicable.

3.2 Bridge wire subassembly, continuity. -- Before assembling the bridge wire subassembly into the ignition element subassembly, it shall be thoroughly cleaned to remove solder flux and contamination. After cleaning, the bridge wire shall be flashed by application of an electric current to assure continuity of the bridge wire. The current shall be regulated to produce a visible red glow of the bridges. An EMF of approximately 1.5 volts ac or dc may be used with a suitable series resistor to adapt the generator characteristics to the primer resistance. Flashing is a necessary manufacturing operation and shall be conducted on each assembly. Assemblies in which the bridge wire fails to glow shall be screened from acceptable material in the process. The discarded assemblies may be reworked by replacement of the bridge.

3.3 Bridge wire resistance. -- The resistance of the bridge wire shall be in accordance with the applicable drawing when tested as specified in 4.5.1.

3.4 Stock extensions. -- Each lot of stock extensions shall pass the stock extension strain test specified in 4.5.2.

3.5 Percussion functioning. -- The primer shall function when the plunger is actuated by a 16.34 ounce steel ball falling from a height of 10.0 inches. Functioning shall not result in a blowback, split stock or detachment of

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metallic components from the primer. The primer shall not function when the plunger is actuated by a 16.34 ounce steel ball falling from a height of 3.0 inches. The test provisions of 4.5.3 shall be applicable in ascertaining conformance.

3.6 Primer sensitivity. -- The sensitivity of the primer shall be ascertained as prescribed in 4.5.4. The sensitivity values shall be used for data card record and process control. Deviations in sensitivity values shall not directly cause the rejection of primers, but may be reason for checking materials and processes for conformance with drawing and specification requirements.

3.7 Electrical functioning. -- Each lot of primers shall pass the electrical functioning test specified in 4.5.5.

3.8 Vibration. -- The primers shall conform to the vibration test as specified in Standard MIL-STD-353 and 4.5.6.

3.9 Jolt test. -- Each primer lot shall pass the jolt test specified in 4.5.7 without firing or being rendered unsafe to handle or transport.

3.10 Ballistic test. -- Each lot of primers shall pass the ballistic test specified in 4.5.8. All primers shall give proper ignition, and there shall be no misfires, hangfires or ruptured stock extensions.

3.11 Primer caps. -- Primer caps used in assembly shall contain none of the defects specified in NAVORD OS 919 and shall be within the specified sensitivity range. The primer caps shall not be deformed in any way during insertion, or in final assembly.

3.11.1 Primer cap seating. -- The depth of the primer cap in the plunger cup shall be gaged prior to assembly of the plunger to insure proper seating of the primer cap.

3.12 Moisture content.

3.12.1 Black powder. -- The moisture content of the black powder used in the stock extension, ignition tube, and ignition cup mixture shall not exceed 0.70 percent.

3.12.2 Gun cotton and pyrocellulose. -- The moisture content of the gun cotton and pyrocellulose shall not exceed 0.50 percent.

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3.13 Primer weighing. -- The Contractor shall check weigh all completed primers to insure presence of black powder charge.

3.14 Cleanliness. -- All component parts shall be thoroughly cleaned and dried before assembly. Operations shall be conducted so as to prevent inclusion of moisture, oil, and foreign matter in the primer. The cleaning methods which are used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent which is used.

3.15 Nonmetallic parts. -- Where parts involved are of plastic or fiber, such parts after final assembly and machining shall be kept in airtight containers until ready for use in final primer assembly.

3.16 Final assembly. -- The methods of loading and assembling shall be uniform for each lot. Shellac used on the ignition tube shall not be used in excess and shall be allowed to dry thoroughly before the ignition tube is filled with black powder. All flash holes shall be completely sealed and the excess lacquer shall be removed from the surface of the stock extension. Lacquer shall be thoroughly dried before the stock extension is used in assembly operations. The stock extension plug shall be dipped in paraffin prior to use.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Sampling. -- Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government Inspector in the determination of the acceptability of products submitted by a supplier for Government Inspection will be in accordance with the provisions of Standard MIL-STD-105.

4.1.1 Lot. -- As applied to the Government Inspection of units of product, the term "Lot" shall mean "Inspection Lot", i.e., a collection of units of the product used as a basis for Government Inspection. Unless otherwise specified, the number of units of product in "Inspection Lots" shall be as determined by the Government Inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.2 Preproduction sample. -- Before entering into quantity production, a preproduction sample of three hundred (300) complete Primers Mk 13 of one specified Mod shall be delivered by the Contractor for test to an agency designated

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by the Procuring Agency. These primers shall be manufactured by the same procedures and processes and the the same location proposed by the Contractor for the execution of the contract. When a preproduction sample is received which proves to be satisfactory for the purpose intended, the Contractor will be notified and will be authorized by the Procuring Agency to proceed with the submission of the product for acceptance. Accepted samples will become the property of the Procuring Agency and will be included in the number called for in the contract schedule.

4.2.1 Preproduction sample for a subsequent contract. -- The necessity for a preproduction sample will be determined by the Procuring Agency when production under a new contract by the same Contractor at the same location follows the manufacture of satisfactory material as covered by this specification.

4.3 Production lot. -- For production purposes the production lot shall consist of 20,000 complete primers of one Mod plus those required for all test purposes. Unless otherwise stated in the contract, order, or requisition, all units for test purposes shall be provided at the expense of the contractor. The production lot should not be confused with the lot for inspection purposes defined in 4.1.1.

4.4 Inspection. -- The general inspection procedures shall be in accordance with "General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment", OP 400; Government Inspection shall be in accordance with the provisions of General Procedures for Government Inspection and Contractor's Inspection shall be in accordance with Inspection by Contractor, as contained in OP 400.

4.5 Tests.

4.5.1 Electrical characteristics. -- Conformance with the bridge circuit resistance of the applicable drawing shall be ascertained by means of a test circuit which limits the bridge current to 50 milliamperes maximum. A safety chamber with an interlock switch shall be used to protect the operator. Primers which conform to the resistance requirements shall be acceptable. The insulator shall be held firmly by the stock and shall not be broken or displaced.

4.5.2 Stock extension strain test. -- Stock extensions shall be subjected to the mercurous nitrate strain test. These stock extensions shall successfully pass an immersion

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in an aqueous solution containing one percent of mercurous nitrate and one percent of nitric acid (specific gravity 1.42) for 15 minutes at room temperature. After 24 hours have elapsed, the mercury shall be removed by volatilization and the samples examined for cracks. Stock extensions which show cracks shall be considered defective. Sampling and disposition of a maximum lot of 25,000 stock extensions shall be in accordance with Standard MIL-STD-105, Table VI-1, Single Sampling Plan, AQL 1.0 percent defective. Stock extensions which have been subjected to this test shall be scrapped.

4.5.3 Percussion functioning test. -- Fit Test Set Mk 135 Mod 0, Bureau of Ordnance List of Drawings LD No. 166537, with a firing chamber and plunger actuator. Mounting of the primer shall simulate support against the forces of explosion equivalent to service operation. Adjust the scale, and align the test set as provided in NAVORD OD 5823 to deliver impacts coincident with the axis of the plunger. Conduct one test for functioning at a drop ball height of 10.0 inches. Defectives for this test include failure to fire, hangfire which is obvious to operator without instrumentation, blowback, split stock, and detached metallic components. Conduct a second test for not functioning at a drop ball height of 3.0 inches in which functioning indicates a defective. Sampling and disposition of the lot on the basis of each individual test shall be in accordance with Standard MIL-STD-105, Inspection Level I, Multiple Sampling Plan, AQL 0.25 percent defective.

4.5.4 Primer sensitivity test. -- Conduct the test on the basis of 50 primers from each production lot. Test the primers in Test Set Mk 135 Mod 0 under mounting provisions of 4.5.3. The staircase method (Bruceton) as prescribed in NAVORD OD 5823 shall be applicable. Use a 16.34 ounce drop ball and 0.50 inch drop increments starting at a suggested height of 6.0 inches for the test. Compute the 50 percent firing height ($\bar{h}$) and the standard deviation (S) and record the values on the data card specified in 5.4.

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4.5.5 Electrical functioning. -- The test shall be conducted in a safety chamber which simulates support against the forces of explosion equivalent to service operation. Fifty (50) of the primers from each production lot shall be subjected to a potential difference of twenty (20) volts d-c, and fifty (50) shall be subjected to a potential difference of two (2) volts d-c. The primer stock and plunger shall be used as terminals. It is important that firm contact on clean bright surfaces is obtained to avoid high contact resistance. All of the primers shall fire, and none of the primers shall suffer damage as evidenced by blowbacks, split stock, or the detachment of metallic components. The firing time of ten (10) primers at twenty (20) volts shall be measured. If the average time of the group of ten (10) primers plus three times the sample standard deviation exceeds fifty (50) milliseconds, the lot shall be rejected.

4.5.6 Vibration test. -- Twenty-four (24) primers from each production lot shall be subjected to the vibration test specified in Standard MIL-STD-353. Rigidly mount the primers to the vibration table. The sample orientations for the three (3) phases of the test shall be axis vertical - stock up, axis vertical - stock down, and axis horizontal. Subject the primers to the vibration schedule of MIL-STD-353. The lot shall be rejected if any of the primers explodes or is unsafe to handle and use for service purposes. The lot shall be rejected if more than one (1) primer fails to meet the bridge resistance requirements of the applicable drawing. Divide the sample into three (3) groups of eight (8) primers each. Subject one (1) group to the electrical functioning test of 4.5.5. Subject the remaining two (2) groups to a percussion functioning test specified in 4.5.3. One (1) group shall be subjected to the ten (10) inch drop, and the other group shall be subjected to the three (3) inch drop. A primer which does not fire at ten (10) inches or a primer which does fire at three (3) inches shall be considered defective. The incidence of more than one (1) defective in the functioning tests shall cause rejection of the lot. The incidence of one (1) defective in the functioning tests shall cause a retest on the basis of a sample of forty-eight (48) primers with rejection of the lot on obtaining more than one (1) defective.

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4.5.7 Jolt test. -- The Inspector shall subject ten primers from each production lot to 600 jolts in each of three positions. The first position shall be vertical with the primer stock up, the second shall be vertical with the primer stock down, the third shall be horizontal. The jolting shall consist of four inch drops upon a leather covered wood block, approximately 1 3/4 inches thick mounted on the iron base of the jolt machine. None of the primers shall fire or be rendered unsafe to handle or transport, as determined by examination.

4.5.8 Ballistic tests. -- Twenty-five of the primers from each production lot shall be assembled into service cartridges. At least five cartridges shall be loaded with proof or near proof powder charges unless it is clearly impractical to do so without special firing in which case all cartridges shall be loaded with service charges. Fifteen primers shall be fired electrically and ten by percussion. Pressure, velocities, and ejection times shall be recorded on at least five rounds fired. All primers shall give proper ignition and there shall be no misfires, hangfires or ruptured stock extensions. If any primer fails to meet these requirements, a retest of the lot may be made. The retest procedure will depend on the nature of the failure and will be specified by the Procuring Agency.

4.5.8.1 Place of test. -- The ballistic tests specified in 4.5.8 and any other test which may be designated by the Procuring Agency will be conducted at a facility designated by the Procuring Agency.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Inert parts. -- Inert parts, including subassemblies, shall be packed in suitable containers so constructed as to insure adequate protection of parts from corrosion or other damage during shipment and storage. If so stipulated in the contract, containers shall be in accordance with any Bureau of Ordnance drawings that may be furnished. Inert parts and subassemblies shall be, at all times, duly segregated by lots to permit ready identification.

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5.1.2 Loaded primers. -- Primers shall be packaged in hermetically sealed containers, similar to the "Primer Container Mk 11 Type". For details of applicable pieces, see Bureau of Ordnance Drawing No. 563645.

5.2 Packing. -- Multiple unit containers shall be packed in accordance with "Primer Container Mk 7 Mod 0 or 1", Bureau of Ordnance List of Drawings LD Nos. 168789 or 257521, respectively.

5.3 Marking. -- In addition to any special marking required by the contract or order, all containers shall be marked in compliance with Standard MIL-STD-129 and Code of Federal Regulations 49 CFR 71-78.

5.4 Data cards. -- Data cards five inches by eight inches, printed as shown below shall be filled out for each lot. One card shall be placed in the top of each container, five in each shipping box, two shall accompany ballistic samples, five shall be forwarded to the Procuring Agency, and twenty shall be forwarded to the Ammunition Depot receiving the primers.

AMMUNITION DATA CARD

Type, Mk and Mod _____ Lot No. _____
Project Order No. _____ Lot Size _____
Loaded by _____ Date _____
Inspected by _____ Commanding Officer _____
Percussion sensitivity $\bar{h}$ _____ s _____

MATERIAL	MANUFACTURER	LOT NO.	YEAR MFG.
Inert parts set			
Black powder FFG			
Primer cap			
Black powder FFFG			

6. NOTES

6.1 Intended use. -- The Primers, Combination, Mk 13 Mods 0, 1, and 2 are combination electric-percussion type primers used to ignite the propellent charges in cartridges for 4-inch, 5-inch and 6-inch case guns.

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